

Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) — YEAR 9 SEMESTER 1

WEEKS 12-18: SOFTWARE, INTERNET, CYBERSECURITY, EMERGING TECH, PROGRAMMING, REVISION & ASSESSMENT

WEEK 12: SOFTWARE & TRANSLATORS

LESSON 12.1 — OPERATING SYSTEM FUNCTIONS

Topic: Software — Operating System Functions

Duration: 40 minutes

Lesson Type: Theory with real-world context

Syllabus Ref: 1.4.1

Learning Objective:

Describe the functions of an operating system; understand memory management, process management, and user interface.

Key Vocabulary:

- operating system (操作系统 / cao zuo xi tong)
- memory management (内存管理 / nei cun guan li)
- process management (进程管理 / jin cheng guan li)
- user interface (用户界面 / yong hu jie mian)
- multitasking (多任务处理 / duo ren wu chu li)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "Your phone is running WeChat, playing music, and downloading a file at the same time."

| "How does your phone manage all of this without crashing?"

3-12 min | DIRECT INSTRUCTION

| Operating system = manager of the computer. Five key functions:

| (1) Memory management — allocates RAM to programs, prevents conflicts.

| (2) Process management — schedules CPU time, multitasking, priority.

| (3) Peripheral management — drivers for keyboard, mouse, printer.

| (4) File management — organises storage, permissions, directories.

| (5) User interface — GUI (Windows, macOS) vs CLI (Terminal, Command Prompt).

| Real-world: iOS vs Android, Windows vs Linux.

12-25 min | GUIDED PRACTICE

- | Scenario cards: "A student opens 5 Chrome tabs, Word, and Spotify."
- | Class identifies which OS function handles each task.
- | Memory management = RAM allocation; Process management = CPU scheduling.

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 12A: Match OS functions to scenarios. Label GUI vs CLI screenshots.
- | Extension: "Why does a game console need a different OS than a smartphone?"

35-40 min | PLENARY + HOMEWORK

- | "If the OS crashes, why does the whole computer stop? What can't work without it?"
- | Homework: List 3 OS functions your phone used today. Identify GUI vs CLI elements.

EAL Support:

- "Manager" analogy: OS = principal of a school managing teachers, rooms, resources
- Bilingual: 操作系统 (operating system), 内存管理 (memory management)
- Visual: screenshots of Windows GUI and Terminal CLI side by side
- Sentence frame: "The operating system manages ____ by ____"

Differentiation:

Emerging | Pre-completed matching activity, focus on 3 core functions

Developing | Standard worksheet, scenario identification

Secure | Extension: Compare mobile OS vs desktop OS functions; research real-time OS

Resources Needed:

- Worksheet 12A
- Screenshots of GUI and CLI interfaces
- Scenario cards (printed)
- Bilingual keyword cards

LESSON 12.2 — TRANSLATORS & IDES

Topic: Translators & IDEs — Compiler, Interpreter, Assembler

Duration: 40 minutes

Lesson Type: Theory with comparison and practical introduction

Syllabus Ref: 1.4.1

Learning Objective:

Compare compilers, interpreters, and assemblers; understand the role of an IDE; introduce Thonny.

Key Vocabulary:

- compiler (编译器 / bian yi qi)
- interpreter (解释器 / jie shi qi)
- assembler (汇编器 / hui bian qi)

- source code (源代码 / yuan dai ma)
- machine code (机器码 / ji qi ma)
- IDE (集成开发环境 / ji cheng kai fa huan jing)
- syntax error (语法错误 / yu fa cuo wu)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "You write an essay in English, but your friend only reads Chinese."
- | "How do you communicate? What are the different ways to translate?"

3-12 min | DIRECT INSTRUCTION

- | Three translators:
- | (1) Compiler — translates entire program at once (C++, Java). Creates executable.
- | Pros: fast execution. Cons: errors found after full compilation.
- | (2) Interpreter — translates line by line (Python, JavaScript). Stops at first error.
- | Pros: immediate feedback. Cons: slower execution.
- | (3) Assembler — translates assembly language (mnemonics like ADD, LOAD) to machine code.
- | IDE: text editor + translator + debugger + run environment. Thonny for Python.
- | Demo: Open Thonny, write `print("Hello VPA")`, run. Show syntax error.

12-25 min | GUIDED PRACTICE

- | Comparison table: compiler vs interpreter. Class fills in speed, error detection, examples.
- | "Which is better for beginners? Why?" (Interpreter — immediate feedback.)

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 12B: Identify translator type from description. Match IDE features to purposes.
- | Extension: "Why can't we just write in machine code?"

35-40 min | PLENARY + HOMEWORK

- | "Python uses an interpreter. What happens if you forget a closing bracket?"
- | "How is this different from a compiled language like C++?"
- | Homework: Install Thonny at home. Write a program that prints your name 3 times.
- | Bring screenshot to next lesson.

EAL Support:

- Translation analogy: compiler = translate entire book before reading; interpreter = live translator, sentence by sentence
- Bilingual: 编译器 (compiler), 解释器 (interpreter), 语法错误 (syntax error)
- Physical: show printed Python code → machine code concept with colour coding
- Sentence frame: "A ___ translates ___ into ___ line by line / all at once"

Differentiation:

Emerging | Focus on concept not technical detail; pre-completed comparison table

Developing | Standard worksheet, Thonny introduction with guided steps

Secure | Research: JIT compilation (Java), transpilers (TypeScript); compare debugging tools

Resources Needed:

- Worksheet 12B
- Thonny IDE installed on all computers
- Projector for live demo
- Bilingual keyword cards

WORKSHEET 12A — OPERATING SYSTEM FUNCTIONS

Name: _____ Class: _____ Date: _____

Learning Objective: Describe the functions of an operating system

SECTION A: OS Functions Match-Up

Match each function to its description:

| Function | Description |

|-----|-----|

| Memory Management | |

| Process Management | |

| Peripheral Management | |

| File Management | |

| User Interface | |

Descriptions:

- a) Organises files and folders, controls access permissions
- b) Allocates RAM to running programs, prevents conflicts
- c) Provides GUI or CLI for user interaction
- d) Manages CPU time between multiple programs
- e) Handles communication with keyboard, mouse, printer

SECTION B: Scenario Identification

For each scenario, identify which OS function is most important:

1. You open Photoshop and it needs 4GB RAM. The OS must ensure other programs still run.

Function: _____

2. You print a document while listening to music and browsing the web.

Function: _____

3. You save your essay in Documents/English/Essay1.docx.

Function: _____

4. 4. You click an icon to open an app.

Function: _____

5. 5. Your game runs smoothly even though 10 apps are open.

Function: _____

SECTION C: GUI vs CLI

6. 6. Look at these two screenshots. Label which is GUI and which is CLI.

GUI: _____ CLI: _____

7. 7. Give two advantages of GUI over CLI:

a) _____

b) _____

8. 8. Give two advantages of CLI over GUI:

a) _____

b) _____

SECTION D: Exam Questions

9. 9. Explain why an operating system needs memory management. [2 marks]

10. 10. Describe two differences between a GUI and a CLI. [2 marks]

Bilingual Support:

operating system = 操作系统 (cao zuo xi tong)

memory management = 内存管理 (nei cun guan li)

process management = 进程管理 (jin cheng guan li)

user interface = 用户界面 (yong hu jie mian)

WORKSHEET 12B — TRANSLATORS & IDES

Name: _____ Class: _____ Date: _____

Learning Objective: Compare compilers, interpreters, and assemblers; understand IDEs

SECTION A: Translator Identification

Read each description and identify the translator type:

11. 1. Translates the entire program before execution. Creates a standalone executable file.

Type: _____

12. 2. Translates one line at a time, executing immediately. Stops at the first error.

Type: _____

13.3. Converts assembly language mnemonics (like ADD, LOAD) into binary machine code.

Type: _____

14.4. Used by Python and JavaScript. Good for beginners because of immediate feedback.

Type: _____

15.5. Used by C++ and Java. Faster execution but slower development cycle.

Type: _____

SECTION B: Compiler vs Interpreter Comparison

Feature	Compiler	Interpreter
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Translation speed	Fast	Slow
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Execution speed	Slow	Fast
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Error detection	At compile time	At run time
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Creates executable?	Yes	No
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Examples	C, C++, Java	Python, JavaScript
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Best for beginners?	No	Yes
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SECTION C: IDE Features

Match each IDE feature to its purpose:

Feature	Purpose
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Syntax highlighting	Makes code easier to read
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Error highlighting	Shows where there is a mistake
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Code completion	Suggests code as you type
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Debugger	Runs the program without leaving the editor
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Run button	Lets you step through code line by line to find bugs
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Purposes:

a) Suggests code as you type, reducing typing errors

b) Shows which line has a mistake before running

c) Runs the program without leaving the editor

d) Different colours for keywords, variables, and comments

e) Lets you step through code line by line to find bugs

SECTION D: Exam Questions

16.6. Explain the difference between a compiler and an interpreter. [3 marks]

17.7. Describe the purpose of an IDE. [2 marks]

18.8. Why is an interpreter often better for learning programming than a compiler? [2 marks]

SECTION E: Thonny Practical

19.9. Open Thonny. Type this code exactly (including the error):

```
print("Hello VPA"
```

What error message do you see? _____

What type of error is this? _____

How would you fix it? _____

20.10. Now type this correct code and run it:

```
print("Hello VPA")
```

```
print("I am learning Python")
```

Screenshot your Thonny window and attach it here.

Bilingual Support:

compiler = 编译器 (bian yi qi)

interpreter = 解释器 (jie shi qi)

assembler = 汇编器 (hui bian qi)

source code = 源代码 (yuan dai ma)

machine code = 机器码 (ji qi ma)

IDE = 集成开发环境 (ji cheng kai fa huan jing)

syntax error = 语法错误 (yu fa cuo wu)

ANSWER KEY — WEEK 12

Worksheet 12A:

SECTION A: Memory Management = b; Process Management = d; Peripheral Management = e;

File Management = a; User Interface = c

SECTION B: 1. Memory Management 2. Process Management 3. File Management

4. User Interface 5. Process Management

SECTION C: 6. (GUI = graphical with icons; CLI = text-based with commands)

- 7. Easier to learn, visual, point-and-click, no need to remember commands
- 8. Faster for experts, uses less memory, powerful for automation/scripts

SECTION D: 9. Prevents programs from using the same memory; ensures RAM is allocated efficiently

- 10. GUI uses icons/windows/pointing device; CLI uses typed commands/text only

Worksheet 12B:

SECTION A: 1. Compiler 2. Interpreter 3. Assembler 4. Interpreter 5. Compiler

SECTION B: Compiler: slow translation, fast execution, errors after full compile, yes creates .exe,

C++/Java, no not best for beginners

Interpreter: fast translation, slower execution, errors immediately, no .exe,

Python/JS, yes best for beginners

SECTION C: Syntax highlighting = d; Error highlighting = b; Code completion = a;

Debugger = e; Run button = c

SECTION D: 6. Compiler translates entire program before execution; interpreter translates line by line;

compiler creates executable; interpreter gives immediate feedback

- 7. IDE combines editor, translator, debugger, and run environment in one program

8. Immediate error feedback; no need to compile entire program; easier to test small changes

SECTION E: 9. SyntaxError: unexpected EOF while parsing (or similar); syntax error; add closing bracket

WEEK 13: INTERNET & NETWORKS

LESSON 13.1 — WWW, URLs, DNS

Topic: Internet — World Wide Web, URLs, DNS

Duration: 40 minutes

Lesson Type: Theory with real-world analogy

Syllabus Ref: 1.5.1

Learning Objective:

Explain the difference between the internet and the World Wide Web; describe URLs and DNS.

Key Vocabulary:

- World Wide Web (万维网 / wan wei wang)
- URL (统一资源定位符 / tong yi zi yuan ding wei fu)
- DNS (域名系统 / yu ming xi tong)
- domain name (域名 / yu ming)
- IP address (IP 地址 / IP di zhi)
- web server (网络服务器 / wang luo fu wu qi)
- web browser (网络浏览器 / wang luo liu lan qi)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "You type 'www.baidu.com' into your browser. What happens in the next 2 seconds?"

| Brainstorm in pairs. No wrong answers.

3-12 min | DIRECT INSTRUCTION

| Internet = global network of connected devices (hardware: cables, routers, satellites).

| World Wide Web = service that runs ON the internet (software: websites, hyperlinks).

| Analogy: Internet = roads; WWW = delivery trucks on those roads.

| URL structure: protocol://domain/path. Example: <https://www.vpa.edu.cn/about>

| DNS = phone book for the internet. Translates domain names to IP addresses.

| Process: browser → DNS server → IP address → web server → web page.

| Demo: nslookup baidu.com (if available) or show DNS lookup diagram.

12-25 min | GUIDED PRACTICE

| Break down URLs: <https://www.bilibili.com/video/BV123456>

| Identify: protocol (https), domain (www.bilibili.com), path (/video/BV123456).

| "Why do we use domain names instead of IP addresses?" (Easier to remember.)

25-35 min | INDEPENDENT PRACTICE

| Worksheet 13A: Label URL parts. Match DNS steps to order. Diagram the request process.

| Extension: "What happens if the DNS server is down?"

35-40 min | PLENARY + HOMEWORK

| "Is WeChat part of the WWW? Why or why not?"

| Homework: Trace the URL of your favourite website. Identify protocol, domain, and path.

| Research: What is your home IP address? (Search 'what is my IP').

EAL Support:

- Phone book analogy for DNS: names (domains) → numbers (IP addresses)
- Bilingual: 万维网 (World Wide Web), 域名系统 (DNS), 网络服务器 (web server)
- Visual: large diagram of DNS lookup process on board
- Sentence frame: "The DNS translates ____ into ____ so that ____"

Differentiation:

Emerging | Pre-labelled URL diagram, focus on concept not technical detail

Developing | Standard worksheet, URL breakdown practice

Secure | Extension: Research DNS caching, CDN, how China's DNS works differently

Resources Needed:

- Worksheet 13A
- Large URL diagram for board
- Internet access for live nslookup demo (optional)
- Bilingual keyword cards

LESSON 13.2 — HTTP, HTTPS & IP ADDRESSES

Topic: Networks — HTTP/HTTPS, IP Addresses

Duration: 40 minutes

Lesson Type: Theory with security focus

Syllabus Ref: 1.5.1

Learning Objective:

Compare HTTP and HTTPS; explain IP addresses; understand why HTTPS matters for security.

Key Vocabulary:

- HTTP (超文本传输协议 / chao wen ben chuan shu xie yi)
- HTTPS (安全超文本传输协议 / an quan chao wen ben chuan shu xie yi)
- IP address (IP 地址 / IP di zhi)
- router (路由器 / lu you qi)
- packet (数据包 / shu ju bao)
- SSL/TLS (安全套接层 / an quan tao jie ceng)

Lesson Structure (40 Minutes):**0-3 min | DO NOW**

| "You see 'http:/' and 'https:/' at the start of web addresses."

| "What is the difference? Why does your bank use https but a blog might use http?"

3-12 min | DIRECT INSTRUCTION

| HTTP = HyperText Transfer Protocol. Unencrypted. Data sent as plain text.

| HTTPS = HTTP Secure. Encrypted using SSL/TLS. Data scrambled.

| Analogy: HTTP = sending a postcard (anyone can read it).

| HTTPS = sending a letter in a locked envelope (only recipient can open).

- | IP address = unique identifier for every device on a network.
- | IPv4: 32-bit (e.g., 192.168.1.1). IPv6: 128-bit (e.g., 2001:0db8::1).
- | Router = directs packets between networks using IP addresses.
- | Why HTTPS matters: passwords, credit cards, personal data protected.

12-25 min | GUIDED PRACTICE

- | Class activity: "Spot the secure site." Show 5 URLs. Students identify HTTP vs HTTPS.
- | Discuss: "Would you enter your password on an HTTP site? Why not?"

25-35 min | INDEPENDENT PRACTICE

| Worksheet 13B: HTTP vs HTTPS comparison. IP address identification. Exam questions.

- | Extension: "Why is IPv6 necessary when we have IPv4?"

35-40 min | PLENARY + HOMEWORK

- | "A website has a green padlock. What does this tell you? Is it 100% safe?"
- | Homework: Check 5 websites you use. How many use HTTPS? Report any that don't.
- | Research: What is a VPN and how does it relate to IP addresses?

EAL Support:

- Postcard vs locked envelope analogy for HTTP vs HTTPS
- Bilingual: 超文本传输协议 (HTTP), 安全超文本传输协议 (HTTPS), 路由器 (router)
- Real-world: show actual browser lock icons, certificate warnings
- Sentence frame: "HTTPS is more secure than HTTP because it ____"

Differentiation:

Emerging | Focus on visual identification (padlock icon), pre-completed comparison table

Developing | Standard worksheet, exam questions on security

Secure | Research: SSL certificates, man-in-the-middle attacks, IPv6 adoption in China

Resources Needed:

- Worksheet 13B
- Browser with example HTTP and HTTPS sites
- Projector for showing lock icons and certificate details
- Bilingual keyword cards

WORKSHEET 13A — WWW, URLs & DNS

Name: _____ Class: _____ Date: _____

Learning Objective: Explain the WWW, URLs, and DNS

SECTION A: Internet vs World Wide Web

21.1. The internet is: _____

22.2. The World Wide Web is: _____

23.3. Complete the analogy: Internet is to roads as WWW is to _____

SECTION B: URL Structure

Break down this URL: <https://www.bilibili.com/video/BV123456>

24.4. Protocol: _____

25.5. Domain name: _____

26.6. Path: _____

27.7. What would the URL be if you wanted the homepage of bilibili.com?

SECTION C: DNS Process

Put these steps in the correct order (1-5):

___ The web server sends the web page back

___ The browser sends a request to the DNS server

___ The user types a URL into the browser

___ The DNS server returns the IP address

___ The browser contacts the web server using the IP address

28.8. Why do we use domain names instead of IP addresses?

SECTION D: Exam Questions

29.9. Explain the difference between the internet and the World Wide Web. [2 marks]

30.10. Describe the role of DNS in accessing a website. [3 marks]

Bilingual Support:

World Wide Web = 万维网 (wan wei wang)

URL = 统一资源定位符 (tong yi zi yuan ding wei fu)

DNS = 域名系统 (yu ming xi tong)

IP address = IP 地址 (IP di zhi)

web server = 网络服务器 (wang luo fu wu qi)

WORKSHEET 13B — HTTP, HTTPS & IP ADDRESSES

Name: _____ Class: _____ Date: _____

Learning Objective: Compare HTTP and HTTPS; understand IP addresses

SECTION A: HTTP vs HTTPS

| Feature | HTTP | HTTPS |

|-----|-----|-----|

| Encrypted? | | |

| Data sent as | | |

| Safe for passwords? | | |

| URL prefix | | |

| Browser icon | | |

| Example use | | |

31. 1. Why is HTTPS essential for online banking? [2 marks]

32. 2. Can a website with HTTPS still be dangerous? Explain. [2 marks]

SECTION B: IP Addresses

33. 3. What is an IP address? _____

34. 4. IPv4 uses ____ bits. Example: _____

35. 5. IPv6 uses ____ bits. Example: _____

36. 6. Why is IPv6 necessary? _____

SECTION C: Router

37. 7. What is the role of a router? _____

38. 8. How does a router use IP addresses? _____

SECTION D: Exam Questions

39. 9. Explain why HTTPS is more secure than HTTP. [3 marks]

40. 10. Describe the difference between IPv4 and IPv6. [2 marks]

Bilingual Support:

HTTP = 超文本传输协议 (chao wen ben chuan shu xie yi)

HTTPS = 安全超文本传输协议 (an quan chao wen ben chuan shu xie yi)

IP address = IP 地址 (IP di zhi)

router = 路由器 (lu you qi)

SSL/TLS = 安全套接层 (an quan tao jie ceng)

ANSWER KEY — WEEK 13

Worksheet 13A:

SECTION A: 1. Global network of connected devices (hardware) 2. Service running on the internet (websites, hyperlinks)

3. delivery trucks

SECTION B: 4. https 5. www.bilibili.com 6. /video/BV123456 7. https://www.bilibili.com

SECTION C: Order: 5, 2, 1, 3, 4

8. Domain names are easier to remember than numbers; IP addresses change

SECTION D: 9. Internet = hardware/network infrastructure; WWW = software service on top

10. DNS translates domain names to IP addresses; browser contacts DNS server;

DNS returns IP; browser uses IP to find web server

Worksheet 13B:

SECTION A: HTTP: No, plain text, No, http://, no padlock/warning, blogs/news

HTTPS: Yes, encrypted, Yes, https://, green padlock, banking/shopping

1. Encrypts data; prevents hackers stealing passwords/account details

2. Yes — HTTPS only encrypts connection; site could still be malicious/phishing

SECTION B: 3. Unique identifier for every device on a network

4. 32, 192.168.1.1 5. 128, 2001:0db8::1

6. IPv4 addresses are running out; IPv6 provides many more addresses

SECTION C: 7. Directs data packets between networks; connects LAN to internet

8. Reads destination IP address; forwards packet to correct network

SECTION D: 9. HTTPS encrypts data using SSL/TLS; prevents eavesdropping;

protects passwords and personal information; shows padlock icon

10. IPv4 = 32-bit, ~4 billion addresses, dotted decimal;

IPv6 = 128-bit, virtually unlimited, hexadecimal with colons

WEEK 14: CYBERSECURITY

LESSON 14.1 — CYBERSECURITY THREATS

Topic: Cybersecurity — Threats (Malware, Phishing, Social Engineering)

Duration: 40 minutes

Lesson Type: Theory with real-world examples

Syllabus Ref: 1.5.2

Learning Objective:

Identify common cybersecurity threats; explain how malware, phishing, and social engineering work.

Key Vocabulary:

- malware (恶意软件 / e yi ruan jian)
- virus (病毒 / bing du)
- worm (蠕虫 / ru chong)
- trojan (木马 / mu ma)
- phishing (网络钓鱼 / wang luo diao yu)
- social engineering (社会工程学 / she hui gong cheng xue)
- ransomware (勒索软件 / le suo ruan jian)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "You receive an email: 'Your package has arrived. Click here to track.'"

| "You didn't order anything. What should you do? Why?"

3-12 min | DIRECT INSTRUCTION

| Cybersecurity = protecting systems, networks, and data from digital attacks.

| Types of malware:

| (1) Virus — attaches to files, spreads when opened. Needs user action.

| (2) Worm — self-replicates across networks. No user action needed.

| (3) Trojan — disguised as legitimate software. Tricks user into installing.

| (4) Ransomware — encrypts files, demands payment for decryption key.

| Phishing = fake emails/websites designed to steal personal information.

| Social engineering = manipulating people into revealing confidential information.

| Real-world: WannaCry ransomware (2017), fake WeChat login pages, phone scams.

12-25 min | GUIDED PRACTICE

| "Spot the threat" activity: Show 5 scenarios (email, popup, USB found, phone call, Wi-Fi).

| Class identifies threat type and response.

| Example: Free USB drive in parking lot = Trojan/ransomware risk.

25-35 min | INDEPENDENT PRACTICE

| Worksheet 14A: Identify threat from description. Design a phishing awareness poster.

| Extension: "Why are humans often the weakest link in cybersecurity?"

35-40 min | PLENARY + HOMEWORK

| "A friend says 'I have antivirus, so I'm completely safe.' Is this true? Why not?"

| Homework: Find one real cybersecurity incident from the news. Identify the threat type.

| Write 3 sentences summarising what happened and how it could have been prevented.

EAL Support:

- Real examples: show sanitised phishing emails, fake popups (printed, not live)
- Bilingual: 恶意软件 (malware), 网络钓鱼 (phishing), 勒索软件 (ransomware)
- Visual: malware type icons (virus = spider, worm = snake, trojan = horse)
- Sentence frame: "This is a ____ threat because it ____"

Differentiation:

Emerging | Focus on recognition and response; pre-completed threat table

Developing | Standard worksheet, scenario analysis

Secure | Research: Zero-day exploits, APTs, state-sponsored attacks; case study analysis

Resources Needed:

- Worksheet 14A
- Sanitised phishing email examples (printed)
- Malware type icons/posters
- Bilingual keyword cards

LESSON 14.2 — CYBERSECURITY PROTECTIONS

Topic: Cybersecurity — Protections (Firewalls, Antivirus, 2FA, Encryption)

Duration: 40 minutes

Lesson Type: Theory with practical application

Syllabus Ref: 1.5.2

Learning Objective:

Explain cybersecurity protections; evaluate which protection is best for different scenarios.

Key Vocabulary:

- firewall (防火墙 / fang huo qiang)
- antivirus (杀毒软件 / sha du ruan jian)
- two-factor authentication (双因素认证 / shuang yin su ren zheng)
- encryption (加密 / jia mi)
- backup (备份 / bei fen)
- patch (补丁 / bu ding)
- penetration testing (渗透测试 / shen tou ce shi)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "You lock your front door, install a security camera, and give spare keys only to family."

| "How is this like cybersecurity? What does each action represent?"

3-12 min | DIRECT INSTRUCTION

| Protections:

| (1) Firewall = barrier between trusted network and untrusted internet.

| Blocks unauthorised access while allowing legitimate traffic.

| (2) Antivirus = scans files for known malware signatures; quarantines threats.

| (3) Two-Factor Authentication (2FA) = password + second factor (SMS, app, fingerprint).

| (4) Encryption = scrambles data so only authorised parties can read it.

| (5) Backup = copies of data stored separately. Ransomware protection.

| (6) Patch/Update = fixes security vulnerabilities. WannaCry spread via unpatched systems.

| Defence in depth = multiple layers of protection. No single solution is enough.

12-25 min | GUIDED PRACTICE

| Scenario: "VPA Shenzhen is building a new computer lab. What protections are needed?"

| Class brainstorm: firewall (network), antivirus (computers), 2FA (teacher accounts),

| encryption (student data), backup (daily), updates (automatic).

25-35 min | INDEPENDENT PRACTICE

| Worksheet 14B: Match protection to threat. Design a security plan for a scenario.

| Extension: "Why is defence in depth better than relying on one strong protection?"

35-40 min | PLENARY + HOMEWORK

| "You forgot your password. The website lets you reset it by answering 'What is your pet's name?'"

| "Is this secure? What could go wrong? How would you improve it?"

| Homework: Enable 2FA on one of your accounts (if possible). Write down which method you used.

| Research: What is biometric authentication? Is it better than passwords?

EAL Support:

- Castle defence analogy: firewall = moat, antivirus = guards, 2FA = double gate, encryption = secret code, backup = escape tunnel
- Bilingual: 防火墙 (firewall), 杀毒软件 (antivirus), 双因素认证 (two-factor authentication)
- Real-world: show 2FA setup on phone, discuss WeChat Pay security
- Sentence frame: "To protect against ____, we should use ____ because ____"

Differentiation:

Emerging | Focus on recognition and basic matching; pre-completed protection table

Developing | Standard worksheet, scenario security planning

Secure | Research: Zero trust architecture, biometric vulnerabilities, quantum encryption

Resources Needed:

- Worksheet 14B
- Phone for 2FA demo (if available)
- Castle defence diagram
- Bilingual keyword cards

WORKSHEET 14A — CYBERSECURITY THREATS

Name: _____ Class: _____ Date: _____

Learning Objective: Identify common cybersecurity threats

SECTION A: Malware Types

Match each malware type to its description:

| Type | Description |

|-----|-----|

| Virus | |

| Worm | |

| Trojan | |

| Ransomware | |

Descriptions:

- a) Disguised as legitimate software; tricks user into installing
- b) Encrypts files and demands payment for the decryption key
- c) Attaches to files; spreads when infected file is opened
- d) Self-replicates across networks without user action

SECTION B: Threat Identification

Identify the threat type for each scenario:

41. 1. You receive an email from "Alipay" asking you to verify your password by clicking a link.

The sender address is al1pay-security@mail.ru

Threat: _____

42. 2. A pop-up says "Your computer is infected! Call this number immediately!"

Threat: _____

43. 3. Your files are suddenly encrypted and a message demands 0.5 Bitcoin to unlock them.

Threat: _____

44. 4. You download a "free game" from an unknown website. Your computer slows down and starts sending emails automatically.

Threat: _____

45. 5. Someone calls pretending to be from your bank and asks for your account details.

Threat: _____

SECTION C: Social Engineering

46. 6. What is social engineering? _____

47. 7. Give two examples of social engineering:

a) _____

b) _____

SECTION D: Exam Questions

48. 8. Explain what is meant by phishing. [2 marks]

49. 9. Describe two differences between a virus and a worm. [2 marks]

50. 10. Explain why ransomware is particularly dangerous for hospitals. [2 marks]

Bilingual Support:

malware = 恶意软件 (e yi ruan jian)

virus = 病毒 (bing du)

worm = 蠕虫 (ru chong)

trojan = 木马 (mu ma)

phishing = 网络钓鱼 (wang luo diao yu)

ransomware = 勒索软件 (le suo ruan jian)

social engineering = 社会工程学 (she hui gong cheng xue)

WORKSHEET 14B — CYBERSECURITY PROTECTIONS

Name: _____ Class: _____ Date: _____

Learning Objective: Explain cybersecurity protections and evaluate their use

SECTION A: Protection Match-Up

Match each protection to the threat it best defends against:

| Protection | Best Against |

|-----|-----|

| Firewall | |

| Antivirus | |

| Two-Factor Authentication | |

| Encryption | |

| Regular Backups | |

| Software Updates | |

Threats:

- a) Stolen password used to access account
- b) Malware sent over the internet
- c) Ransomware encrypting all files
- d) Unauthorised access to network from outside
- e) Data intercepted during transmission
- f) Exploits in old software versions

SECTION B: Defence in Depth

51. 1. What is meant by "defence in depth"? _____

52. 2. Why is defence in depth better than using only one protection? _____

SECTION C: Scenario — VPA Shenzhen Computer Lab

Victoria Park Academy Shenzhen is setting up a new computer lab with 30 student PCs,

a teacher workstation, and a school network. Design a security plan.

53. 3. Network protection: _____

54. 4. Computer protection: _____

55.5. Account protection: _____

56.6. Data protection: _____

57.7. What is the weakest link in most security systems? _____

SECTION D: Exam Questions

58.8. Explain the purpose of a firewall. [2 marks]

59.9. Describe how two-factor authentication improves security. [2 marks]

60.10. Explain why regular software updates are important for cybersecurity. [2 marks]

Bilingual Support:

firewall = 防火墙 (fang huo qiang)

antivirus = 杀毒软件 (sha du ruan jian)

two-factor authentication = 双因素认证 (shuang yin su ren zheng)

encryption = 加密 (jia mi)

backup = 备份 (bei fen)

patch = 补丁 (bu ding)

ANSWER KEY — WEEK 14

Worksheet 14A:

SECTION A: Virus = c; Worm = d; Trojan = a; Ransomware = b

SECTION B: 1. Phishing 2. Scareware/Social engineering 3. Ransomware

4. Trojan/Worm 5. Social engineering

SECTION C: 6. Manipulating people into revealing confidential information or performing actions

7. Pretexting, baiting, quid pro quo, tailgating (any two valid)

SECTION D: 8. Fake emails/websites designed to trick users into revealing personal information

- 9. Virus needs user action to spread; worm self-replicates automatically;
virus attaches to files; worm spreads across networks
- 10. Locks patient records; delays treatment; could endanger lives; hospitals pay quickly

Worksheet 14B:

SECTION A: Firewall = d; Antivirus = b; 2FA = a; Encryption = e;

Backups = c; Updates = f

SECTION B: 1. Multiple layers of security; if one fails, others still protect

- 2. No single protection is perfect; layered defence reduces risk

SECTION C: 3. Firewall to block unauthorised access; network segmentation

- 4. Antivirus on all PCs; restricted software installation
- 5. Strong passwords + 2FA for teacher accounts; limited student privileges
- 6. Encrypted storage for student data; regular backups to separate location
- 7. Humans / users (social engineering, weak passwords, clicking links)

SECTION D: 8. Monitors incoming/outgoing network traffic; blocks unauthorised access;

- allows legitimate traffic; acts as barrier
- 9. Requires two different proofs of identity; even if password stolen,
attacker needs second factor; significantly reduces unauthorised access
- 10. Fixes security vulnerabilities; patches known exploits;
WannaCry spread via unpatched systems; keeps protection up to date

WEEK 15: EMERGING TECHNOLOGIES

LESSON 15.1 — DIGITAL CURRENCY & BLOCKCHAIN

Topic: Emerging Tech — Digital Currency and Blockchain

Duration: 40 minutes

Lesson Type: Theory with real-world context

Syllabus Ref: 1.6.1

Learning Objective:

Explain digital currency and blockchain; understand advantages and risks of decentralised systems.

Key Vocabulary:

- digital currency (数字货币 / shu zi huo bi)
- blockchain (区块链 / qu kuai lian)
- decentralised (去中心化 / qu zhong xin hua)
- cryptocurrency (加密货币 / jia mi huo bi)
- Bitcoin (比特币 / bi te bi)
- ledger (账本 / zhang ben)
- mining (挖矿 / wa kuang)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "You send 100 RMB to a friend using WeChat Pay. Where does the money go?"
- | "Who keeps track of the transaction? What if WeChat's servers crash?"

3-12 min | DIRECT INSTRUCTION

- | Digital currency = money that exists only in digital form. No physical coins/notes.
- | Examples: WeChat Pay balance, Alipay, Bitcoin, central bank digital currency (e-CNY).
- | Blockchain = distributed digital ledger. Transactions grouped in "blocks," linked in a "chain."
- | Key features: decentralised (no single bank controls it), transparent (all can see),
- | immutable (cannot be changed once recorded).
- | How it works: transaction → verified by network (mining) → added to block → block added to chain.
- | Real-world: China's e-CNY (digital yuan), Bitcoin, supply chain tracking.
- | Risks: price volatility, energy consumption (mining), regulatory uncertainty, scams.

12-25 min | GUIDED PRACTICE

- | Analogy activity: "The class ledger." Each student = node. One student writes a transaction.
- | All students copy it. If someone tries to change it, everyone notices. This is blockchain.
- | Compare: traditional bank (one ledger, one authority) vs blockchain (many ledgers, many authorities).

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 15A: Blockchain diagram labelling. Digital currency comparison. Exam questions.
- | Extension: "Why does China ban Bitcoin mining but promote e-CNY?"

35-40 min | PLENARY + HOMEWORK

- | "Your grandmother asks: 'Is Bitcoin real money?' How do you answer?"
- | Homework: Research China's e-CNY (数字人民币). How is it different from Bitcoin?
- | Write 3 bullet points comparing the two.

EAL Support:

- Ledger analogy: blockchain = class diary where everyone has a copy; no one can secretly change it
- Bilingual: 数字货币 (digital currency), 区块链 (blockchain), 去中心化 (decentralised)
- Visual: diagram of blockchain structure (blocks linked by hashes)
- Sentence frame: "Blockchain is different from a traditional bank because it is ____"

Differentiation:

Emerging | Focus on concept and analogy; pre-completed blockchain diagram

Developing | Standard worksheet, comparison activities

Secure | Research: Smart contracts, DeFi, NFTs, environmental impact of mining

Resources Needed:

- Worksheet 15A
- Class ledger activity materials (paper, pens)
- Blockchain diagram for board
- Bilingual keyword cards

LESSON 15.2 — ARTIFICIAL INTELLIGENCE & ROBOTICS

Topic: Emerging Tech — AI, Machine Learning, and Robotics

Duration: 40 minutes

Lesson Type: Theory with ethical discussion

Syllabus Ref: 1.6.1

Learning Objective:

Explain artificial intelligence and machine learning; discuss benefits and ethical concerns of AI and robotics.

Key Vocabulary:

- artificial intelligence (人工智能 / ren gong zhi neng)
- machine learning (机器学习 / ji qi xue xi)
- robotics (机器人技术 / ji qi ren ji shu)
- algorithm (算法 / suan fa)
- training data (训练数据 / xun lian shu ju)
- bias (偏见 / pian jian)
- automation (自动化 / zi dong hua)

Lesson Structure (40 Minutes):**0-3 min | DO NOW**

- | "Your phone suggests the next word when you type a message."
- | "Your map app predicts traffic. Your music app recommends songs."
- | "What do these have in common? Is this 'intelligence'?"

3-12 min | DIRECT INSTRUCTION

| Artificial Intelligence (AI) = computer systems that can perform tasks requiring human intelligence.

| Machine Learning (ML) = subset of AI where systems learn from data without explicit programming.

| How ML works: training data → algorithm learns patterns → makes predictions on new data.

| Examples: facial recognition, voice assistants (Siri, Xiao Ai), recommendation algorithms,

| medical diagnosis, autonomous vehicles.

| Robotics = design and use of robots. Physical AI. Examples: factory arms, drones, surgical robots.

| Benefits: efficiency, accuracy, dangerous tasks, 24/7 operation.

| Concerns: job displacement, bias in algorithms, privacy, autonomous weapons, over-reliance.

| Real-world: DeepSeek, ChatGPT, Tesla Autopilot, factory automation in Shenzhen.

12-25 min | GUIDED PRACTICE

| "AI or not AI?" activity: Show 10 scenarios. Class categorises.

| Calculator = not AI. Chess computer = AI (if it learns). Spam filter = ML.

| Discuss: "Is a thermostat AI?" (No — rule-based, not learning.)

25-35 min | INDEPENDENT PRACTICE

| Worksheet 15B: AI vs ML vs robotics identification. Ethical scenario analysis.

| Extension: "Design an AI system for VPA Shenzhen. What data would it need? What could go wrong?"

35-40 min | PLENARY + HOMEWORK

| "A company uses AI to screen job applications. It rejects more women than men."

| "What went wrong? How could this have been prevented?"

| Homework: Find one example of AI bias in the news. Explain what happened and how it could be fixed.

EAL Support:

- "Learning from experience" analogy: ML = student who practises many maths problems and gets better over time; traditional programming = student who only follows a formula book
- Bilingual: 人工智能 (artificial intelligence), 机器学习 (machine learning), 偏见 (bias)
- Visual: flowchart showing AI → ML → Deep Learning hierarchy
- Sentence frame: "Machine learning is different from traditional programming because it ____"

Differentiation:

Emerging | Focus on recognition and basic concepts; pre-completed AI/ML table

Resources Needed:

- Worksheet 15B
- AI/ML scenario cards
- Flowchart diagram for board
- Bilingual keyword cards

WORKSHEET 15A — DIGITAL CURRENCY & BLOCKCHAIN

Name: _____ Class: _____ Date: _____

Learning Objective: Explain digital currency and blockchain

SECTION A: Digital Currency Basics

61. 1. What is digital currency? _____

62. 2. Give three examples of digital currency:
a) _____ b) _____ c) _____
63. 3. How is digital currency different from physical cash? _____

SECTION B: Blockchain

64. 4. What is a blockchain? _____

65. 5. Label this diagram:
[Block 1] → [Block 2] → [Block 3] → [Block 4]
Each block contains:
- _____ (list of transactions)
- _____ (link to previous block)
- _____ (unique identifier)
66. 6. Why is blockchain called "decentralised"? _____

67. 7. What makes blockchain transactions immutable (cannot be changed)?

SECTION C: Comparison

68. 8. Complete the table:
- | Feature | Traditional Banking | Blockchain |
|------------------|---------------------|------------|
| Who controls it? | | |

Who can see transactions?		
Can transactions be changed?		
Speed		
Example		

SECTION D: Exam Questions

69. 9. Explain what is meant by blockchain. [2 marks]

70. 10. Describe two advantages of using blockchain for financial transactions. [2 marks]

71. 11. Describe two risks of using cryptocurrency. [2 marks]

Bilingual Support:

digital currency = 数字货币 (shu zi huo bi)

blockchain = 区块链 (qu kuai lian)

decentralised = 去中心化 (qu zhong xin hua)

cryptocurrency = 加密货币 (jia mi huo bi)

Bitcoin = 比特币 (bi te bi)

ledger = 账本 (zhang ben)

WORKSHEET 15B — AI, MACHINE LEARNING & ROBOTICS

Name: _____ **Class:** _____ **Date:** _____

Learning Objective: Explain AI, ML, and robotics; discuss ethical concerns

SECTION A: Definitions

72. 1. Artificial Intelligence is: _____

73. 2. Machine Learning is: _____

74. 3. Robotics is: _____

SECTION B: AI or Not AI?

Identify whether each example uses AI (yes/no) and explain:

75. 4. A calculator that adds two numbers: _____

Explanation: _____

76.5. A spam filter that learns which emails you delete: _____

Explanation: _____

77.6. A robot vacuum that maps your house and avoids obstacles: _____

Explanation: _____

78.7. A washing machine that runs for 30 minutes at 40C: _____

Explanation: _____

SECTION C: Machine Learning Process

79.8. Put these steps in order (1-4):

___ The system makes predictions on new data

___ The algorithm identifies patterns in the training data

___ Collect large amounts of training data

___ The system is tested and refined

SECTION D: Ethical Concerns

80.9. Give two benefits of AI in healthcare:

a) _____

b) _____

81.10. Give two ethical concerns about AI:

a) _____

b) _____

82.11. What is "algorithmic bias"? _____

SECTION E: Exam Questions

83.12. Explain what is meant by machine learning. [2 marks]

84.13. Describe two benefits of using robots in manufacturing. [2 marks]

85.14. Explain why bias in training data is a problem for AI systems. [3 marks]

Bilingual Support:

artificial intelligence = 人工智能 (ren gong zhi neng)

machine learning = 机器学习 (ji qi xue xi)

robotics = 机器人技术 (ji qi ren ji shu)

algorithm = 算法 (suan fa)

training data = 训练数据 (xun lian shu ju)

bias = 偏见 (pian jian)

automation = 自动化 (zi dong hua)

ANSWER KEY — WEEK 15

Worksheet 15A:

SECTION A: 1. Money that exists only in digital/electronic form; no physical coins or notes

2. WeChat Pay, Alipay, Bitcoin, e-CNY (any three)
3. No physical form; stored electronically; transferred digitally

SECTION B: 4. Distributed digital ledger of transactions grouped in blocks linked together

5. Transactions, hash (link to previous block), timestamp/nonce
6. No single authority controls it; maintained by network of computers
7. Each block contains hash of previous block; changing one block breaks the chain;
all nodes would detect the change

SECTION C: Traditional: Bank controls, bank/customer only, yes can be reversed, slow (days), Bank transfer

Blockchain: Network controls, all can see (transparent), no immutable, fast (minutes), Bitcoin

SECTION D: 9. Distributed digital ledger; transactions in blocks linked by cryptography;

- maintained by network of computers
10. Transparent (all can verify); immutable (cannot be altered); no single point of failure;
faster international transfers; lower fees (any two)
 11. Price volatility; energy consumption (mining); regulatory uncertainty; scams/theft;
irreversible transactions (any two)

Worksheet 15B:

SECTION A: 1. Computer systems that can perform tasks requiring human intelligence

2. Subset of AI where systems learn from data without being explicitly programmed
3. Design and use of robots; physical machines that perform tasks automatically

SECTION B: 4. No — follows fixed rules, no learning or adaptation

- 5. Yes — learns from user behaviour (which emails marked as spam)
- 6. Yes — uses sensors and algorithms to map and navigate; may learn over time
- 7. No — follows fixed timer and temperature settings; no learning or decision-making

SECTION C: Order: 3, 2, 4, 1 (Collect data → Identify patterns → Test and refine → Make predictions)

SECTION D: 9. Faster diagnosis; more accurate detection of diseases; 24/7 monitoring;

- access to remote areas (any two)
- 10. Job displacement; privacy concerns; bias in algorithms; autonomous weapons; over-reliance on technology (any two)
- 11. When AI system produces unfair results because training data contains human prejudices

SECTION E: 12. Systems learn from data without explicit programming; identify patterns;

- improve performance with more data
- 13. Work 24/7 without breaks; more precise/consistent; handle dangerous tasks; faster production; lower long-term costs (any two)
- 14. AI learns patterns from training data; if data contains biases, AI replicates them; leads to unfair/discriminatory outcomes; affects decisions in hiring/lending/policing; difficult to detect once system is deployed

WEEK 16: PROGRAMMING INTRODUCTION

LESSON 16.1 — PYTHON BASICS

Topic: Programming Introduction — Python Basics

Duration: 40 minutes

Lesson Type: Practical with theory foundation

Syllabus Ref: 2.1.1

Learning Objective:

Write simple Python programs using print, comments, and basic syntax; understand the programming cycle.

Key Vocabulary:

- Python (Python / Python)
- syntax (语法 / yu fa)

- print statement (打印语句 / da yin yu ju)
- comment (注释 / zhu shi)
- string (字符串 / zi fu chuan)
- integer (整数 / zheng shu)
- debugging (调试 / tiao shi)
- algorithm (算法 / suan fa)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "A recipe tells you exactly how to make a cake. What happens if you miss a step?"
- | "How is a computer program like a recipe? What happens if you make a syntax error?"

3-12 min | DIRECT INSTRUCTION

- | Programming = writing instructions for a computer. Python = high-level, interpreted language.
- | The programming cycle: understand problem → plan algorithm → write code → test → debug → refine.
- | Basic syntax rules:
 - | (1) print() outputs text. Text inside quotes: print("Hello World")
 - | (2) Comments start with # — ignored by computer, for humans only.
 - | (3) Python is case-sensitive: Print and print are different.
 - | (4) Indentation matters in Python (preview for next year).
- | Live demo in Thonny:
 - | print("Hello VPA Shenzhen")
 - | print(5 + 3)
 - | # This is a comment
 - | print("My name is", "Teacher")

12-25 min | GUIDED PRACTICE

- | "Code along" activity: All students type the same 5 lines in Thonny.
- | Teacher projects code. Students run and observe output.
- | Introduce common errors: missing quote, wrong case, missing bracket.

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 16A: Predict output, fix errors, write simple programs.
- | Extension: "Write a program that prints a short poem with 3 lines."

35-40 min | PLENARY + HOMEWORK

- | "You wrote print(Hello) without quotes. What happens? Why?"
- | Homework: Write a Python program that prints your name, age, and favourite subject.

| Each on a separate line. Screenshot the output and bring to next lesson.

EAL Support:

- Recipe analogy: algorithm = recipe; syntax = grammar of the recipe; bug = mistake in recipe
- Bilingual: 语法 (syntax), 打印语句 (print statement), 注释 (comment)
- Visual: project code live; use colour-coded syntax highlighting
- Sentence frame: "To print text, I use ____ with ____ inside the brackets"

Differentiation:

Emerging | Code along with teacher; pre-written code to modify; focus on running programs

Developing | Standard worksheet, write simple programs independently

Secure | Extension: ASCII art with print statements; multi-line output; escape characters

Resources Needed:

- Worksheet 16A
- Thonny IDE on all computers
- Projector for live coding
- Bilingual keyword cards

LESSON 16.2 — VARIABLES, INPUT & OUTPUT

Topic: Programming — Variables, Input/Output

Duration: 40 minutes

Lesson Type: Practical programming

Syllabus Ref: 2.1.1

Learning Objective:

Use variables to store data; write programs that accept user input and produce output.

Key Vocabulary:

- variable (变量 / bian liang)
- assignment (赋值 / fu zhi)
- input (输入 / shu ru)
- output (输出 / shu chu)
- data type (数据类型 / shu ju lei xing)
- string (字符串 / zi fu chuan)
- integer (整数 / zheng shu)
- concatenation (连接 / lian jie)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "You have a box. You write 'age' on it and put the number 15 inside."

| "Later, you change the number to 16. What is this like in programming?"

3-12 min | DIRECT INSTRUCTION

- | Variable = named storage location in memory. Like a labelled box.
- | Assignment: name = value. Example: age = 15
- | Rules: no spaces, no starting with number, case-sensitive.
- | Input: name = input("What is your name? ") — pauses program, waits for user.
- | Output: print("Hello", name) — displays result.
- | Data types: string (text), integer (whole number), float (decimal).
- | Type conversion: int() and str() for combining types.
- | Live demo:
 - | name = input("Enter your name: ")
 - | age = int(input("Enter your age: "))
 - | print("Hello", name + "! You are", age, "years old.")

12-25 min | GUIDED PRACTICE

- | Code along: "Age calculator." Input birth year, calculate age (2026 - birth_year).
- | Common error: forgetting int() conversion. Show the error, fix it together.

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 16B: Variable declaration, input/output programs, error fixing.
- | Extension: "Write a program that asks for two numbers and prints their sum."

35-40 min | PLENARY + HOMEWORK

- | "Why does age = input("Age: ") + 5 cause an error? How do you fix it?"
- | Homework: Write a "Mad Libs" program. Ask for 3 words (noun, adjective, verb).
- | Print a funny sentence using all three words.

EAL Support:

- Box analogy: variable = box with label; assignment = putting something in the box; input = asking someone what to put in the box; output = showing what's inside
- Bilingual: 变量 (variable), 输入 (input), 输出 (output), 数据类型 (data type)
- Visual: draw memory boxes on board showing variable names and values
- Sentence frame: "I store ____ in a variable called ____ using ____"

Differentiation:

Emerging | Pre-written code with blanks to fill; focus on running and modifying

Developing | Standard worksheet, write programs from specifications

Secure | Extension: Multiple variables, formatted output (f-strings preview), simple calculations

Resources Needed:

- Worksheet 16B
- Thonny IDE on all computers

- Projector for live coding
- Bilingual keyword cards

WORKSHEET 16A — PYTHON BASICS

Name: _____ Class: _____ Date: _____

Learning Objective: Write simple Python programs using print and comments

SECTION A: Predict the Output

Predict what each program will print:

86. 1. `print("Hello World")`

Output: _____

87. 2. `print(7 + 5)`

Output: _____

88. 3. `print("7 + 5")`

Output: _____

89. 4. `print("Hello", "VPA")`

Output: _____

90. 5. `print(10 - 3)`

Output: _____

SECTION B: Fix the Errors

Each program has a syntax error. Find and correct it:

91. 6. `Print("Hello")`

Error: _____ Fix: _____

92. 7. `print(Hello)`

Error: _____ Fix: _____

93. 8. `print("Hello)`

Error: _____ Fix: _____

94. 9. `print("Hello" + 5)`

Error: _____ Fix: _____

SECTION C: Write Your Own

95. 10. Write a program that prints your name and your favourite subject on the same line.

96. 11. Write a program that prints this pattern using three print statements:

```
*
**
***
```

97.12. Add a comment to explain what your program in Q10 does.

SECTION D: Challenge

98.13. Write a program that prints:

```
=====
Welcome to VPA Computer Science
=====
```

Bilingual Support:

Python = Python

syntax = 语法 (yu fa)

print statement = 打印语句 (da yin yu ju)

comment = 注释 (zhu shi)

string = 字符串 (zi fu chuan)

integer = 整数 (zheng shu)

WORKSHEET 16B — VARIABLES, INPUT & OUTPUT

Name: _____ Class: _____ Date: _____

Learning Objective: Use variables, input, and output in Python

SECTION A: Variable Basics

99.1. What is a variable? _____

100. 2. Which of these are valid variable names? Circle yes or no:

a) myName Yes / No

b) 2ndScore Yes / No

c) total_mark Yes / No

d) class Yes / No

e) \$price Yes / No

101. 3. What will be stored in each variable after these assignments?

a) age = 15 age = _____

b) name = "VPA" name = _____

c) score = 95.5 score = _____

SECTION B: Input and Output

102. 4. What does this program do?

```
name = input("What is your name? ")  
print("Hello", name)
```

103. 5. What is wrong with this program?

```
age = input("Enter your age: ")  
print("Next year you will be", age + 1)
```

Fix: _____

SECTION C: Write Programs

104. 6. Write a program that asks for the user's favourite colour and prints:

"Your favourite colour is [colour]!"

105. 7. Write a program that asks for two numbers and prints their sum.

106. 8. Write a program that asks for the user's birth year and calculates their age in 2026.

SECTION D: Challenge

107. 9. Write a program that asks for:

- The user's first name
- The user's last name
- Their favourite number

Then prints: "Hello [first] [last]! Your favourite number squared is [result]."

Bilingual Support:

variable = 变量 (bian liang)

assignment = 赋值 (fu zhi)

input = 输入 (shu ru)

output = 输出 (shu chu)

data type = 数据类型 (shu ju lei xing)

string = 字符串 (zi fu chuan)

integer = 整数 (zheng shu)

ANSWER KEY — WEEK 16

Worksheet 16A:

SECTION A: 1. Hello World 2. 12 3. 7 + 5 4. Hello VPA 5. 7

SECTION B: 6. Print should be print (case sensitive)

7. Hello needs quotes: `print("Hello")`

8. Missing closing quote: `print("Hello")`

9. Cannot concatenate string and integer: `print("Hello" + str(5))` or `print("Hello", 5)`

SECTION C: 10. `print("My name is [name] and I like [subject]")`

11. `print("*") / print("") / print("**")`

12. `#` This program prints my name and favourite subject

SECTION D: 13. `print("=====")`

`print("Welcome to VPA Computer Science")`

`print("=====")`

Worksheet 16B:

SECTION A: 1. A named storage location in memory that holds a value

2. a) Yes b) No (starts with number) c) Yes d) Yes (but avoid — reserved word)

e) No (\$ not allowed)

3. a) 15 b) "VPA" c) 95.5

SECTION B: 4. Asks for user's name, then prints a greeting with that name

5. input returns a string; cannot add 1 to string; need `int(age) + 1`

Fix: `age = int(input("Enter your age: "))`

SECTION C: 6. `colour = input("What is your favourite colour? ")`

```
print("Your favourite colour is", colour + "!")
```

7. `num1 = int(input("Enter first number: "))`

```
num2 = int(input("Enter second number: "))
```

```
print("The sum is", num1 + num2)
```

8. `birth_year = int(input("Enter your birth year: "))`

```
age = 2026 - birth_year
```

```
print("You will be", age, "years old in 2026.")
```

SECTION D: 9. `first = input("First name: ")`

```
last = input("Last name: ")
```

```
num = int(input("Favourite number: "))
```

```
print("Hello", first, last + "! Your favourite number squared is", num * num)
```

WEEK 17: REVISION CLINIC

LESSON 17.1 — REVISION: BINARY, HEX & HARDWARE

Topic: Revision Clinic — Data Representation & Hardware

Duration: 40 minutes

Lesson Type: Active revision with retrieval practice

Syllabus Ref: 1.1.1–1.3.3 (Review)

Learning Objective:

Retrieve and apply knowledge from Weeks 1–11; identify gaps in understanding.

Key Vocabulary:

- retrieval practice (检索练习 / jian suo lian xi)
- spaced repetition (间隔重复 / jian ge chong fu)
- gap analysis (差距分析 / cha ju fen xi)
- mock exam (模拟考试 / mo ni kao shi)

Lesson Structure (40 Minutes):

0-3 min | **DO NOW**

| Retrieval quiz (5 questions, no notes):

| 1. Convert 10101101 to denary.

| 2. What is the largest number in 8 bits?

- | 3. Name three CPU registers and their functions.
- | 4. What is the difference between RAM and ROM?
- | 5. Explain packet switching in one sentence.

3-12 min | DIRECT INSTRUCTION

| Revision strategy: active recall > re-reading. Use flashcards, teach someone else, practice questions.

| Quick-fire topic review: binary/hex conversion (Weeks 1–3), data representation (Weeks 4–6),

| hardware (Weeks 9–11). Focus on common errors:

- | - Forgetting to add leading zeros in binary (e.g., 5 = 00000101, not 101)
- | - Confusing MAR and MDR
- | - Forgetting that ROM is non-volatile

12-25 min | GUIDED PRACTICE

| "Revision relay": Teams of 4. Each member answers one question from a different topic.

| Rotate until all 4 topics covered. Fastest correct team wins.

| Topics: (1) Binary/hex conversion, (2) Data representation (images/sound/text),

| (3) CPU/FDE cycle, (4) Memory/storage.

25-35 min | INDEPENDENT PRACTICE

| Worksheet 17A: Mixed revision questions covering Weeks 1–11.

| Students self-mark with answer key. Circle questions they got wrong for gap analysis.

35-40 min | PLENARY + HOMEWORK

| "Which topic do you feel most confident about? Which needs more work?"

| Create a personal revision plan: 2 topics to review this week, 2 resources to use.

| Homework: Complete 10 flashcards for your weakest topic. Bring to next lesson for peer testing.

EAL Support:

- Bilingual flashcards encouraged — English term on one side, Chinese + pinyin on the other
- Sentence frames for extended answers reviewed and displayed
- Pair revision: explain concept to partner in either language, then translate

Differentiation:

Emerging | Focus on core concepts (binary conversion, RAM vs ROM); guided revision with teacher

Developing | Standard revision worksheet, peer testing

Secure | Create revision resources for peers; lead revision relay team

Resources Needed:

- Worksheet 17A
- Flashcard templates (bilingual)
- Answer key for self-marking
- Revision relay question cards

LESSON 17.2 — REVISION: NETWORKS, SOFTWARE & SECURITY

Topic: Revision Clinic — Networks, Software, Cybersecurity, Emerging Tech

Duration: 40 minutes

Lesson Type: Active revision with exam practice

Syllabus Ref: 1.4.1–1.6.1 (Review)

Learning Objective:

Retrieve and apply knowledge from Weeks 12–15; practice exam-style questions.

Lesson Structure (40 Minutes):**0-3 min | DO NOW**

- | Retrieval quiz (5 questions, no notes):
- | 1. What is the difference between a compiler and an interpreter?
- | 2. What does DNS stand for and what does it do?
- | 3. Name two cybersecurity threats and one protection for each.
- | 4. What is blockchain and why is it called "decentralised"?
- | 5. What is the difference between AI and machine learning?

3-12 min | DIRECT INSTRUCTION

- | Exam technique focus: command words (explain, describe, compare, evaluate).
- | "Explain" = give reasons why. "Describe" = say what it is/how it works.
- | "Compare" = identify similarities and differences. "Evaluate" = give pros and cons + judgement.
- | Common errors in written answers: too short, missing technical terms, no examples.
- | Model answer structure: definition + explanation + example.

12-25 min | GUIDED PRACTICE

- | "Exam speed dating": Students rotate every 3 minutes. Each station has one exam question.
- | Write a bullet-point answer. Partner gives feedback using mark scheme.
- | Questions cover: OS functions, HTTP vs HTTPS, malware types, blockchain, AI ethics.

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 17B: Exam-style questions from Weeks 12–15. Timed practice (20 minutes).

| Self-mark with mark scheme. Identify command words and where marks are awarded.

35-40 min | PLENARY + HOMEWORK

| "What is the most important thing to remember when answering a 4-mark 'explain' question?"

| (Answer: at least 4 distinct points; use technical terms; give examples.)

| Homework: Write one perfect exam answer (4 marks) on your weakest topic.

| Use the model structure: definition + explanation + example + link back to question.

EAL Support:

- Command word glossary displayed in English and Chinese
- Model answers provided with colour-coded structure (definition = yellow, explanation = blue, example = green)
- Sentence starters for each command word

Differentiation:

Emerging | Focus on 2-mark questions; sentence frames provided; teacher-guided practice

Developing | Standard exam questions, peer feedback

Secure | 4-mark and 6-mark questions; create mark schemes for peers; evaluate model answers

Resources Needed:

- Worksheet 17B
- Mark schemes
- Exam speed dating question cards
- Command word posters (bilingual)

WORKSHEET 17A — REVISION: BINARY, HEX & HARDWARE

Name: _____ **Class:** _____ **Date:** _____

Learning Objective: Retrieve knowledge from Weeks 1–11

SECTION A: Binary & Hex (Weeks 1–3)

108. 1. Convert 75 to 8-bit binary: _____
109. 2. Convert 10110100 to denary: _____
110. 3. Convert 255 to hexadecimal: _____
111. 4. Convert 1A (hex) to denary: _____
112. 5. Add in binary: 01101010 + 00010111 = _____
113. 6. Does 200 + 100 cause overflow in 8 bits? _____ Explain: _____
- _____

SECTION B: Data Representation (Weeks 4–6)

114. 7. How many bits to store the letter 'A' in ASCII? _____
115. 8. Why is Unicode needed instead of just ASCII? _____
- _____

116. 9. An image is 800 x 600 pixels with 24-bit colour depth. Calculate file size in bytes:

117. 10. What is the difference between lossy and lossless compression?

SECTION C: CPU & FDE Cycle (Week 9)

118. 11. What does the Program Counter (PC) hold? _____

119. 12. What happens during the FETCH stage? _____

120. 13. What happens during the DECODE stage? _____

121. 14. What happens during the EXECUTE stage? _____

122. 15. Why is the PC incremented during the fetch stage? _____

SECTION D: Memory & Storage (Weeks 10–11)

123. 16. Is RAM volatile or non-volatile? _____ What happens when power is lost?

124. 17. What is stored in ROM? _____

125. 18. Why is cache faster than RAM? _____

126. 19. What is virtual memory? _____

127. 20. Give one advantage of SSD over HDD: _____

SECTION E: Self-Assessment

Circle your confidence for each topic (1 = low, 5 = high):

Binary/Hex: 1 2 3 4 5

Data Representation: 1 2 3 4 5

CPU/FDE: 1 2 3 4 5

Memory/Storage: 1 2 3 4 5

Which topic needs the most revision? _____

Bilingual Support:

retrieval practice = 检索练习 (jian suo lian xi)

spaced repetition = 间隔重复 (jian ge chong fu)

gap analysis = 差距分析 (cha ju fen xi)

WORKSHEET 17B — REVISION: NETWORKS, SOFTWARE & SECURITY

Name: _____ Class: _____ Date: _____

Learning Objective: Retrieve knowledge from Weeks 12–15; practice exam technique

SECTION A: Software & Translators (Week 12)

128. 1. Give two functions of an operating system. [2 marks]

129. 2. Explain the difference between a compiler and an interpreter. [3 marks]

SECTION B: Internet & Networks (Week 13)

130. 3. Describe the role of DNS in accessing a website. [3 marks]

131. 4. Explain why HTTPS is more secure than HTTP. [3 marks]

SECTION C: Cybersecurity (Week 14)

132. 5. Describe two types of malware and how they differ. [4 marks]

133. 6. Explain how two-factor authentication improves security. [2 marks]

SECTION D: Emerging Tech (Week 15)

134. 7. Explain what is meant by blockchain. [2 marks]

135. 8. Describe two benefits and two concerns of using AI in healthcare. [4 marks]

SECTION E: Exam Technique Checklist

Before submitting, check:

- ☐ I used the command word correctly (explain/describe/compare/evaluate)
- ☐ I included at least one technical term per mark
- ☐ I gave an example where possible
- ☐ My answer is not too short (aim for 2-3 lines per mark)
- ☐ I checked my spelling of key terms

Bilingual Support:

mock exam = 模拟考试 (mo ni kao shi)

command word = 指令词 (zhi ling ci)

mark scheme = 评分标准 (ping fen biao zhun)

ANSWER KEY — WEEK 17

Worksheet 17A:

SECTION A: 1. 01001011 2. 180 3. FF 4. 26 5. 01111111 (127)

- 6. Yes — result is 300, which exceeds 255 (max for 8 bits)

SECTION B: 7. 8 bits (1 byte) 8. ASCII only supports 128 characters; Unicode supports all languages including Chinese

- 9. $800 \times 600 \times 24 / 8 = 1,440,000$ bytes (or show working)
- 10. Lossy = permanent data loss, smaller files, cannot recover original (e.g., JPEG);
Lossless = no data loss, larger files, fully recoverable (e.g., PNG, RLE)

SECTION C: 11. Address of the next instruction to be fetched

- 12. PC → MAR → Memory → MDR → CIR; PC increments
- 13. CU interprets the instruction in CIR
- 14. ALU performs the operation (arithmetic/logic)
- 15. So the CPU knows where to fetch the next instruction; prevents infinite loop

SECTION D: 16. Volatile; all data lost 17. BIOS/boot instructions (permanent, non-volatile)

- 18. Closer to CPU; faster memory technology; stores frequently used data

- 19. Using hard disk space as RAM when physical RAM is full
- 20. No moving parts (more durable); faster read/write; uses less power; quieter (any one)

Worksheet 17B:

SECTION A: 1. Any two: memory management, process management, peripheral management,

file management, user interface

- 2. Compiler translates entire program before execution; creates executable; interpreter translates line by line; stops at first error; compiler = faster execution; interpreter = immediate feedback

SECTION B: 3. DNS translates domain names to IP addresses; browser sends request to DNS server;

DNS returns IP address; browser uses IP to contact web server

- 4. HTTPS encrypts data using SSL/TLS; HTTP sends data as plain text; HTTPS prevents eavesdropping; protects passwords and personal data; HTTPS shows padlock icon

SECTION C: 5. Any two malware types with differences (see Week 14)

- 6. Requires two proofs of identity; even if password stolen, attacker needs second factor; significantly reduces unauthorised access

SECTION D: 7. Distributed digital ledger; transactions in blocks linked by cryptography;

maintained by network; no central authority

- 8. Benefits: faster diagnosis, more accurate, 24/7 monitoring, access to remote areas (any two)
Concerns: bias in training data, privacy, over-reliance, job displacement (any two)

WEEK 18: MOCK EXAM & REVIEW

LESSON 18.1 — MOCK EXAM (PAPER 1 STYLE)

Topic: Assessment — End-of-Semester Mock Exam

Duration: 40 minutes

Lesson Type: Formal assessment

Syllabus Ref: 1.1.1–1.6.1, 2.1.1 (Full Semester)

Learning Objective:

Demonstrate understanding of all Year 9 Semester 1 topics under exam conditions.

Instructions for Students:

- This is a 40-minute mock exam in the style of CIE IGCSE Computer Science Paper 1.
- Calculators are NOT permitted.
- Write your name, class, and date clearly.
- Read all questions carefully before answering.
- Show your working for calculation questions.
- Use technical terms where appropriate.
- Check your answers if you finish early.

Exam Structure (40 Minutes):

SECTION A: Multiple Choice (10 marks — 10 questions, 1 mark each)

Covers: binary/hex, data representation, hardware, software, networks, security

SECTION B: Short Answer (15 marks — 5 questions, 3 marks each)

Covers: binary conversion, CPU registers, memory hierarchy, packet switching, cybersecurity

SECTION C: Extended Answer (10 marks — 2 questions, 5 marks each)

Covers: data representation (images/sound), emerging technology (AI/blockchain)

Total: 35 marks (scaled to fit 40 minutes; full Paper 1 is 75 marks in 105 minutes)

MOCK EXAM PAPER

Name: _____ Class: _____ Date: _____

Time: 40 minutes

Total Marks: 35

Calculators NOT permitted

SECTION A: MULTIPLE CHOICE (10 marks)

Choose the correct answer for each question.

136. 1. What is the denary value of the binary number 10101010?
A) 160 B) 170 C) 180 D) 190
137. 2. Which of the following is a function of an operating system?
A) Writing programs
B) Memory management
C) Designing websites
D) Creating databases
138. 3. What is the largest positive integer that can be stored in an 8-bit register?
A) 128 B) 255 C) 256 D) 512
139. 4. During the fetch-decode-execute cycle, which register holds the address of the next instruction?

- A) ACC B) MDR C) PC D) CIR
140. 5. Which type of memory is non-volatile and stores boot instructions?
A) RAM B) ROM C) Cache D) Virtual memory
141. 6. What does DNS stand for?
A) Digital Network System
B) Domain Name System
C) Data Network Service
D) Digital Name Server
142. 7. Which protocol is used for secure communication between a browser and a web server?
A) HTTP B) FTP C) HTTPS D) SMTP
143. 8. What is the main advantage of packet switching over circuit switching?
A) Dedicated line for each user
B) More efficient use of network bandwidth
C) Faster for voice calls
D) Simpler to implement
144. 9. Which of the following is an example of social engineering?
A) A virus that self-replicates
B) A fake email asking for your password
C) A firewall blocking unauthorised access
D) An antivirus scanning for malware
145. 10. What is the purpose of a compiler?
A) To execute code line by line
B) To translate source code into machine code all at once
C) To debug programs automatically
D) To manage computer memory

SECTION B: SHORT ANSWER (15 marks)

Answer all questions. Show your working where appropriate.

146. 11. Convert the denary number 200 to 8-bit binary. [2 marks]

147. 12. Explain what is meant by overflow in binary addition. Give an example. [3 marks]

148. 13. Describe the role of the MAR and MDR in the fetch-decode-execute cycle. [3 marks]

149. 14. Explain why cache memory improves CPU performance. [3 marks]

150. 15. Describe two ways a user can protect themselves from phishing attacks. [2 marks]

151. 16. Explain the difference between lossy and lossless compression. [2 marks]

SECTION C: EXTENDED ANSWER (10 marks)

Answer both questions.

152. 17. An image is 1024 pixels wide and 768 pixels high. It uses 24-bit colour depth.

(a) Calculate the file size of the uncompressed image in bytes. Show your working. [3 marks]

(b) Explain why the actual file size might be smaller than your answer. [2 marks]

153. 18. Artificial Intelligence (AI) is being used in many areas of modern life.

(a) Explain what is meant by machine learning. [2 marks]

(b) Describe two benefits of using AI in healthcare. [2 marks]

(c) Describe one ethical concern about the use of AI in decision-making. [1 mark]

END OF EXAM

LESSON 18.2 — MOCK EXAM REVIEW & GAP ANALYSIS

Topic: Mock Review — Feedback and Next Steps

Duration: 40 minutes

Lesson Type: Feedback and target setting

Syllabus Ref: N/A (Assessment Review)

Learning Objective:

Analyse mock exam performance; identify strengths and gaps; set targets for Year 10.

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| "Without looking at your paper, write down 3 things you think you did well and 3 things you found difficult."

3-12 min | DIRECT INSTRUCTION

- | Return marked papers. Explain mark scheme for each section.
- | Common errors to highlight (based on typical class performance):
 - | - Binary conversion: forgetting leading zeros
 - | - FDE cycle: confusing MAR/MDR or forgetting PC increment
 - | - Extended answers: not enough detail, missing examples
 - | - Exam technique: not reading questions carefully, running out of time
- | Grade boundaries (for reference):
 - | A* = 80%+, A = 70%+, B = 60%+, C = 50%+, D = 40%+

12-25 min | GUIDED PRACTICE

- | "Fix it" activity: Students work in pairs on one question they both got wrong.
- | Use mark scheme to identify missing marks. Rewrite answer together.
- | Teacher circulates, provides feedback on common errors.

25-35 min | INDEPENDENT PRACTICE

- | Personal gap analysis worksheet: Students complete target sheet.
- | Identify: (1) strongest topic, (2) weakest topic, (3) one exam technique to improve,
- | (4) one resource to use over the break, (5) one question type to practise.

35-40 min | PLENARY + HOMEWORK

- | "What is one thing you will do differently in Year 10 to improve your Computer Science grade?"
- | Share one target with the class.
- | Homework: Over the break, complete 5 exam questions from your weakest topic.
- | Use the mark scheme to self-assess. Bring to first lesson of Year 10.

EAL Support:

- Mark scheme provided in simplified English with Chinese keywords
- Peer support: stronger students explain in either language
- Visual: colour-coded mark scheme (1 mark = one distinct point)
- Sentence frame: "I need to improve ____ because I lost marks on ____"

Differentiation:

Emerging | Focus on Section A (multiple choice) and one Section B question;
teacher-led review of weakest topic

Developing | Standard gap analysis, peer feedback, target setting

Secure | Lead peer review sessions; create revision guides for weaker topics;
analyse mark schemes to predict examiner expectations

Resources Needed:

- Marked mock exam papers
- Mark scheme (printed)
- Gap analysis/target sheet
- Mark scheme posters (bilingual)

WORKSHEET 18A — MOCK EXAM REVIEW & GAP ANALYSIS

Name: _____ **Class:** _____ **Date:** _____

Mock Exam Score: _____ / 35

Percentage: _____%

SECTION A: Performance Analysis

For each section, write your score and one strength/one area for improvement:

Multiple Choice: _____ / 10

Strength: _____

Improvement: _____

Short Answer: _____ / 15

Strength: _____

Improvement: _____

Extended Answer: _____ / 10

Strength: _____

Improvement: _____

SECTION B: Topic Breakdown

Rate your confidence for each topic (1 = low, 5 = high):

Binary & Hex: 1 2 3 4 5

Data Representation: 1 2 3 4 5

CPU & FDE Cycle: 1 2 3 4 5

Memory & Storage: 1 2 3 4 5
Software & Translators: 1 2 3 4 5
Internet & Networks: 1 2 3 4 5
Cybersecurity: 1 2 3 4 5
Emerging Tech: 1 2 3 4 5

SECTION C: Target Setting

My strongest topic: _____
My weakest topic: _____
One exam technique to improve: _____

One resource I will use: _____

One question type to practise: _____

My target grade for Year 10: _____

SECTION D: Reflection

Write 3 sentences about what you will do differently in Year 10:

154. 1. _____
155. 2. _____
156. 3. _____

Bilingual Support:

gap analysis = 差距分析 (cha ju fen xi)

target setting = 目标设定 (mu biao she ding)

mark scheme = 评分标准 (ping fen biao zhun)

exam technique = 考试技巧 (kao shi ji qiao)

ANSWER KEY — WEEK 18 (MOCK EXAM)

SECTION A: Multiple Choice

157. 1. B (170) 2. B (Memory management) 3. B (255) 4. C (PC)
158. 5. B (ROM) 6. B (Domain Name System) 7. C (HTTPS) 8. B (More efficient use of bandwidth)
159. 9. B (Fake email asking for password) 10. B (Translate source code into machine code all at once)

SECTION B: Short Answer

160. 11. 11001000 ($128 + 64 + 8 = 200$) [2 marks: 1 for correct binary, 1 for showing working]

161. 12. Overflow occurs when a calculation result exceeds the maximum value that can be stored [1];
in 8 bits, maximum is 255 [1]; example: $200 + 100 = 300$, which cannot fit in 8 bits [1]
162. 13. MAR holds the memory address being accessed [1]; MDR holds the data/instruction being transferred [1];
during fetch, MAR receives address from PC, MDR receives instruction from memory [1]
163. 14. Cache stores frequently accessed data closer to CPU [1]; reduces need to access slower RAM [1];
speeds up repeated operations and improves overall performance [1]
164. 15. Any two: check sender address carefully; hover over links before clicking; use 2FA;
never share passwords via email; verify requests through another channel [2 marks, 1 each]
165. 16. Lossy = permanent data loss, smaller files, cannot recover original [1];
Lossless = no data loss, fully recoverable, larger files [1]

SECTION C: Extended Answer

166. 17. (a) $1024 \times 768 \times 24 = 18,874,368$ bits [1]; $/ 8 = 2,359,296$ bytes [1]; show working [1]
(b) Compression reduces file size [1]; metadata may be stored separately [1]
167. 18. (a) Machine learning is a subset of AI where systems learn from data without explicit programming [1];
identify patterns and improve with more data [1]
(b) Any two: faster diagnosis; more accurate detection; 24/7 monitoring; access to remote areas [2]
(c) Any one: bias in training data; privacy concerns; job displacement; over-reliance; lack of transparency [1]

END OF FILE 4: WEEKS 12-18